

BIRZAI, Lithuania

WMO: 265310

Lat: 56.1931N

Lon: 24.7742E

Elev: 202

StdP: 14.59

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.3	3.2	-7.7	3.7	-2.6	-1.6	5.1	4.3	19.8	31.4	18.1	28.6	4.0	110	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	83.1	68.2	79.8	66.2	76.8	64.5	70.1	79.3	68.1	76.6	66.2	74.6	7.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.8	99.5	75.1	64.9	93.0	72.4	63.1	87.0	70.4	34.2	79.2	32.5	76.4	31.0	74.7	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.9	15.0	13.4	DB	-9.8	87.9	7.5	3.5	-15.2	90.4	-19.6	92.4	-23.8	94.3	-29.3	96.9	(4)
(5)			WB	-10.2	72.9	7.4	2.6	-15.5	74.8	-19.8	76.3	-23.9	77.8	-29.3	79.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.7	24.8	25.4	32.7	44.8	54.3	60.5	65.1	63.3	54.8	44.3	35.9	28.7	(6)	
	DBStd	16.29	10.64	10.94	7.92	7.90	7.50	6.02	5.49	5.38	5.89	6.84	7.84	9.57	(7)	
	HDD50	3547	780	690	536	194	40	1	0	0	22	200	424	660	(8)	
	HDD65	7577	1245	1110	1000	605	342	162	68	98	308	640	874	1125	(9)	
	CDD50	1597	0	0	0	39	172	316	467	413	166	24	0	0	(10)	
	CDD65	154	0	0	0	0	9	27	70	46	2	0	0	0	(11)	
	CDH74	1310	0	0	0	6	82	237	587	376	22	0	0	0	(12)	
	CDH80	266	0	0	0	0	7	40	144	74	1	0	0	0	(13)	
(14) Wind	WSAvg	6.6	7.5	7.2	7.2	6.7	6.4	6.1	5.7	5.6	6.0	6.8	7.3	7.5	(14)	
(15) Precipitation	PrecAvg	26.0	1.8	1.6	1.5	1.5	2.1	2.9	3.1	2.7	2.0	2.5	2.1	2.0	(15)	
	PrecMax	34.6	4.1	2.6	2.8	2.8	4.4	5.2	7.7	5.2	4.1	4.6	3.8	2.9	(16)	
	PrecMin	19.8	0.6	0.7	0.3	0.3	0.5	0.6	0.2	0.4	0.4	0.6	0.1	1.2	(17)	
	PrecStd	3.7	0.7	0.6	0.7	0.7	1.0	1.1	1.7	1.4	0.9	1.0	0.9	0.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.0	44.1	57.5	74.5	80.6	84.5	88.6	86.5	77.4	63.9	52.1	48.0	(19)	
		MCWB	43.2	40.7	46.3	57.3	66.1	67.7	69.8	70.5	62.9	56.6	49.8	46.3	(20)	
	2%	DB	41.3	41.6	51.1	68.6	76.1	80.1	84.0	81.7	71.9	59.4	49.4	44.6	(21)	
		MCWB	39.7	39.2	44.1	53.5	62.0	66.2	69.1	66.8	61.0	54.9	47.5	42.9	(22)	
	5%	DB	38.5	39.5	46.2	63.5	72.4	76.4	80.4	78.2	67.9	56.5	47.5	42.0	(23)	
		MCWB	37.1	37.7	41.1	51.4	59.4	64.0	67.3	65.1	59.4	53.4	45.8	40.7	(24)	
	10%	DB	36.3	37.6	42.7	58.7	68.5	72.7	77.0	74.7	64.7	54.1	45.7	39.7	(25)	
		MCWB	35.2	36.2	39.1	49.1	57.3	61.9	65.6	63.4	58.0	51.3	44.3	38.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.8	41.6	48.0	58.5	67.9	71.0	73.1	72.6	65.0	58.1	50.3	46.7	(27)	
		MCDB	44.7	43.1	55.5	71.1	77.8	80.0	83.4	81.9	73.3	61.8	51.5	48.0	(28)	
	2%	WB	39.9	39.6	44.5	55.2	64.3	67.6	70.8	69.1	62.4	55.7	47.9	43.0	(29)	
		MCDB	41.0	41.0	49.9	65.7	72.5	76.9	80.4	77.5	68.5	58.4	49.2	44.3	(30)	
	5%	WB	37.3	38.0	41.9	52.8	61.2	65.4	68.8	66.9	60.5	53.6	46.1	40.8	(31)	
		MCDB	38.4	39.4	45.6	61.4	69.8	74.2	77.5	75.3	65.8	56.2	47.3	41.9	(32)	
	10%	WB	35.4	36.3	39.5	50.1	58.6	63.2	66.9	65.1	58.8	51.8	44.4	38.6	(33)	
		MCDB	36.3	37.5	42.4	57.1	66.0	70.6	74.8	72.7	63.8	53.9	45.5	39.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	9.2	11.7	16.5	18.4	16.7	17.0	17.4	14.8	10.7	6.4	6.4	(35)	
		MCDBR	6.6	7.6	16.9	24.3	23.5	22.2	22.8	23.1	20.2	13.2	7.9	6.3	(36)	
	5% WB	MCWBR	6.2	6.3	11.0	12.0	11.6	10.5	9.8	10.0	10.8	9.1	6.8	6.0	(37)	
		MCWBR	6.2	7.1	15.0	21.1	20.5	19.6	19.9	19.8	16.9	12.1	7.4	6.1	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.290	0.322	0.355	0.395	0.385	0.379	0.412	0.409	0.373	0.348	0.332	0.291	(40)	
		taud	2.175	2.162	2.203	2.231	2.320	2.366	2.294	2.299	2.379	2.399	2.258	2.189	(41)	
	Ebn at Noon	Ebn at Noon	188	228	252	260	271	274	261	253	246	222	171	156	(42)	
		Edh at Noon	18	27	34	38	37	36	38	36	29	22	17	14	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	156	389	808	1255	1685	1751	1656	1347	899	435	145	95	(44)	
	RadStd	RadStd	38	80	135	156	159	153	150	130	98	74	23	16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)			+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page